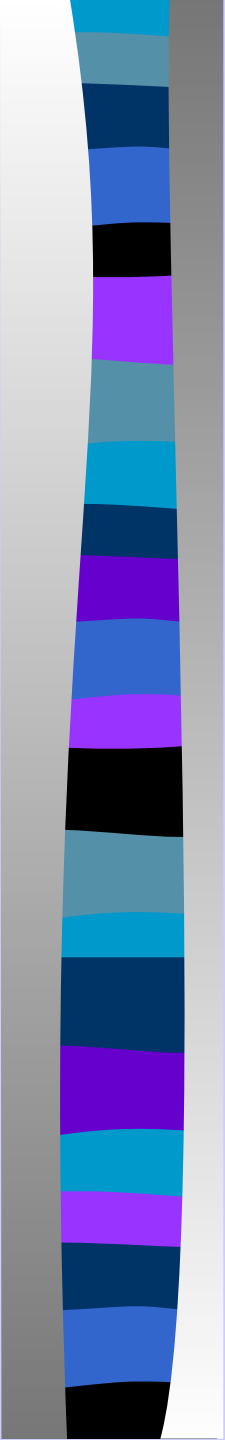


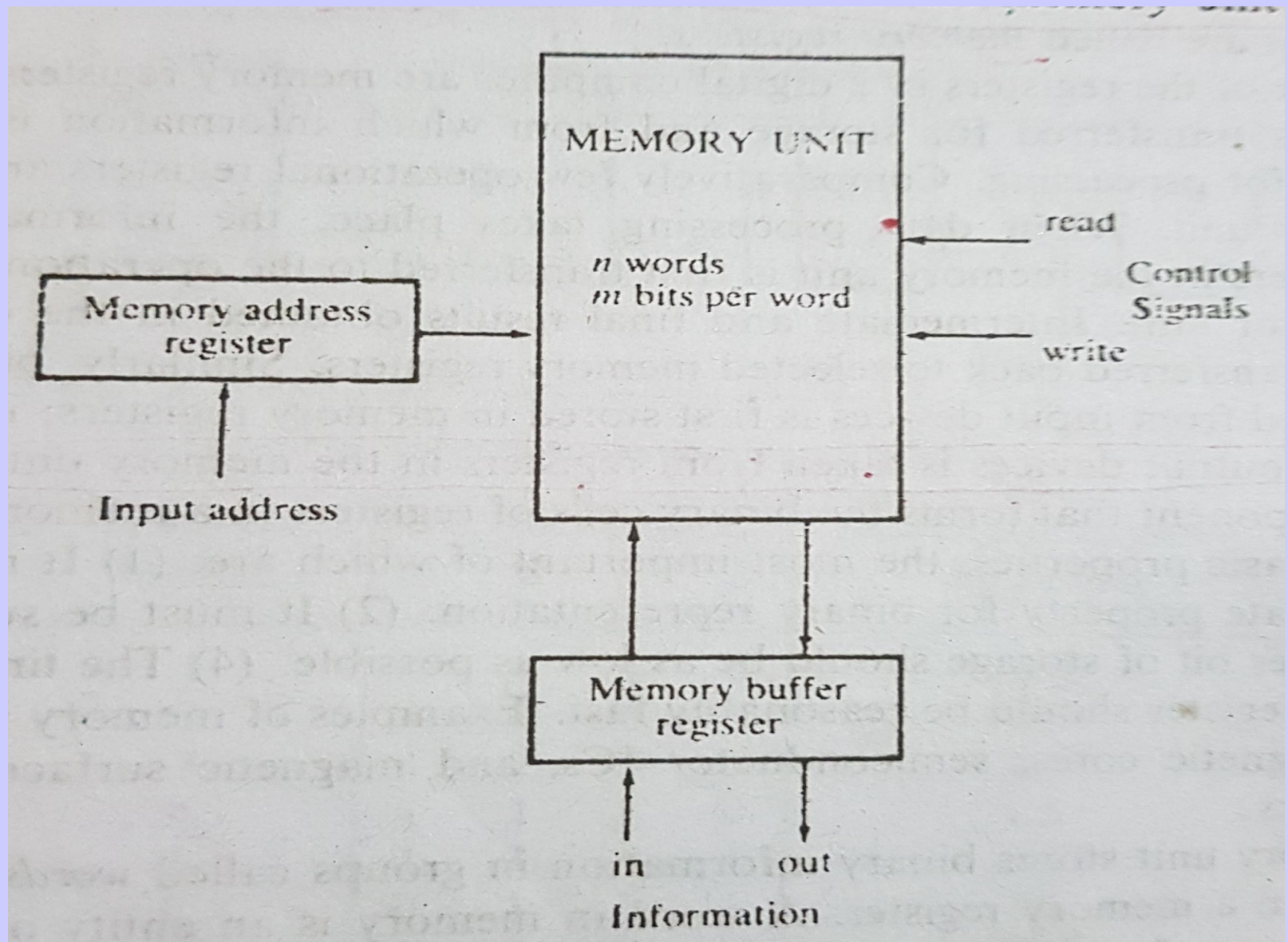
Memory

BRAC University

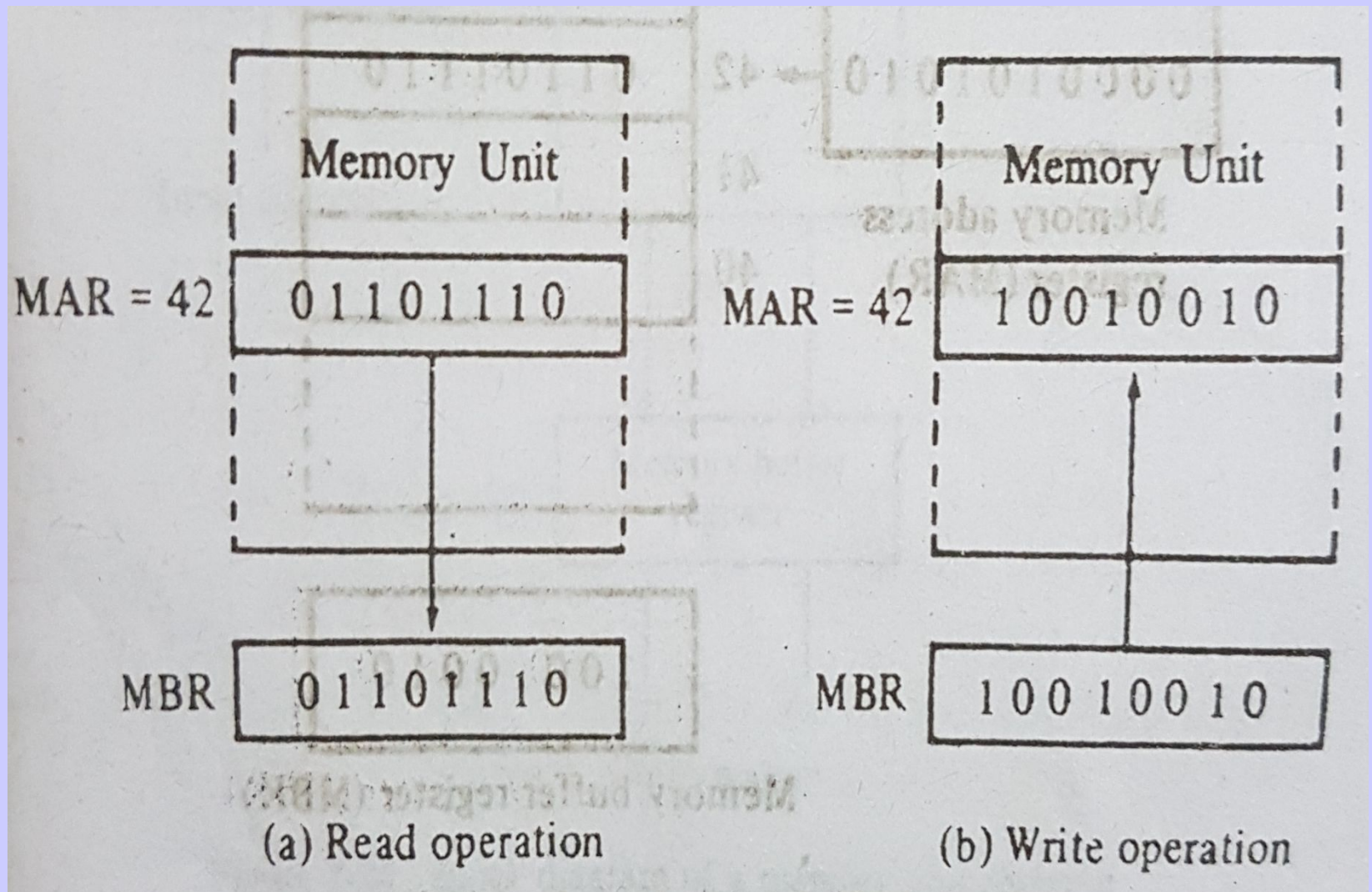


What is Memory Unit?

- A **memory unit** is a **collection of registers together** with associated circuits needed to transfer information in and out of registers.
- **Binary cell:** An elementary unit of **computer storage** (i.e. memory register) that can have one or the other of **two stable states** and can thus **store one bit of information**.
- The memory cell is **an electronic circuit that stores one bit of binary information** and it must be set to store a logic 1 (high voltage level) and reset to store a logic 0 (low voltage level). Its value is maintained/stored until it is changed by the set/reset process.
- A memory unit stores binary information in **groups of bits** called *words*.
- Communication between memory unit and its environment is achieved through **2 controls**(read and write) and **2 external registers** (memory address register and memory buffer register).
- To communicate with a memory word, its address is transferred to **memory address register**. Internal circuit then opens the path to **select** that word. A n-bit address register can specify up to 2^n address. (note: # of address=# of word)
- Before writing and after reading information are kept in **Memory Buffer registers**. In each case, the address register specifies location of writing or reading.



READ WRITE Operation Example



Random Access Memory (RAM)

- Content of a 1024 x 16-bit memory:

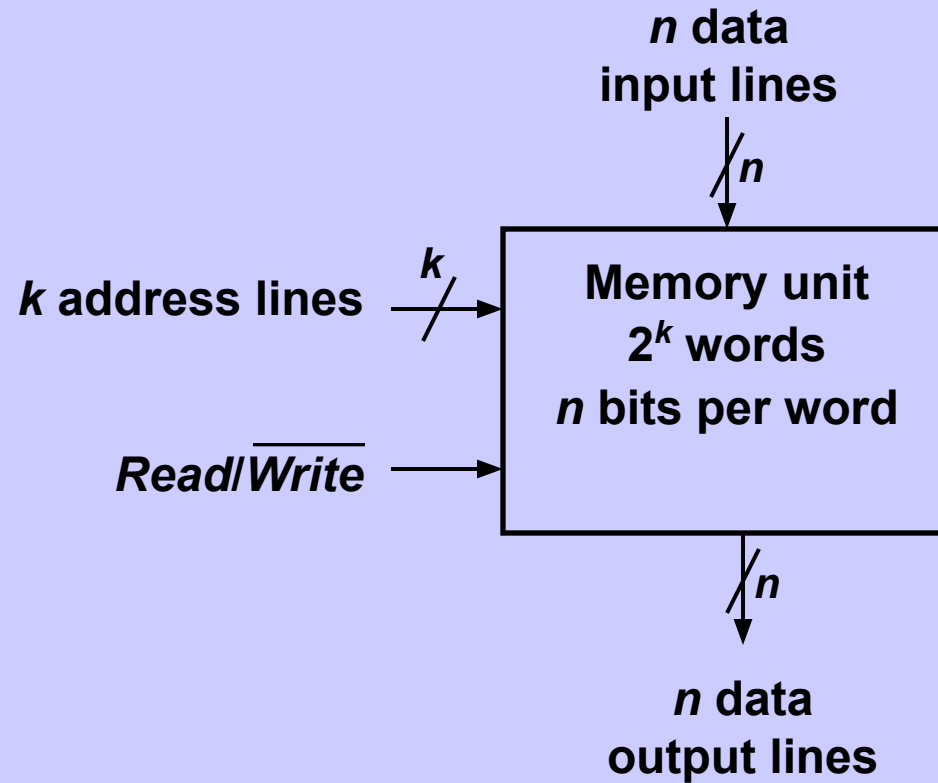
Memory address		Memory content
binary	decimal	
0000000000	0	1011010111011101
0000000001	1	1010000110000110
0000000010	2	0010011101110001
:	:	:
:	:	:
1111111101	1021	1110010101010010
1111111110	1022	0011111010101110
1111111111	1023	1011000110010101

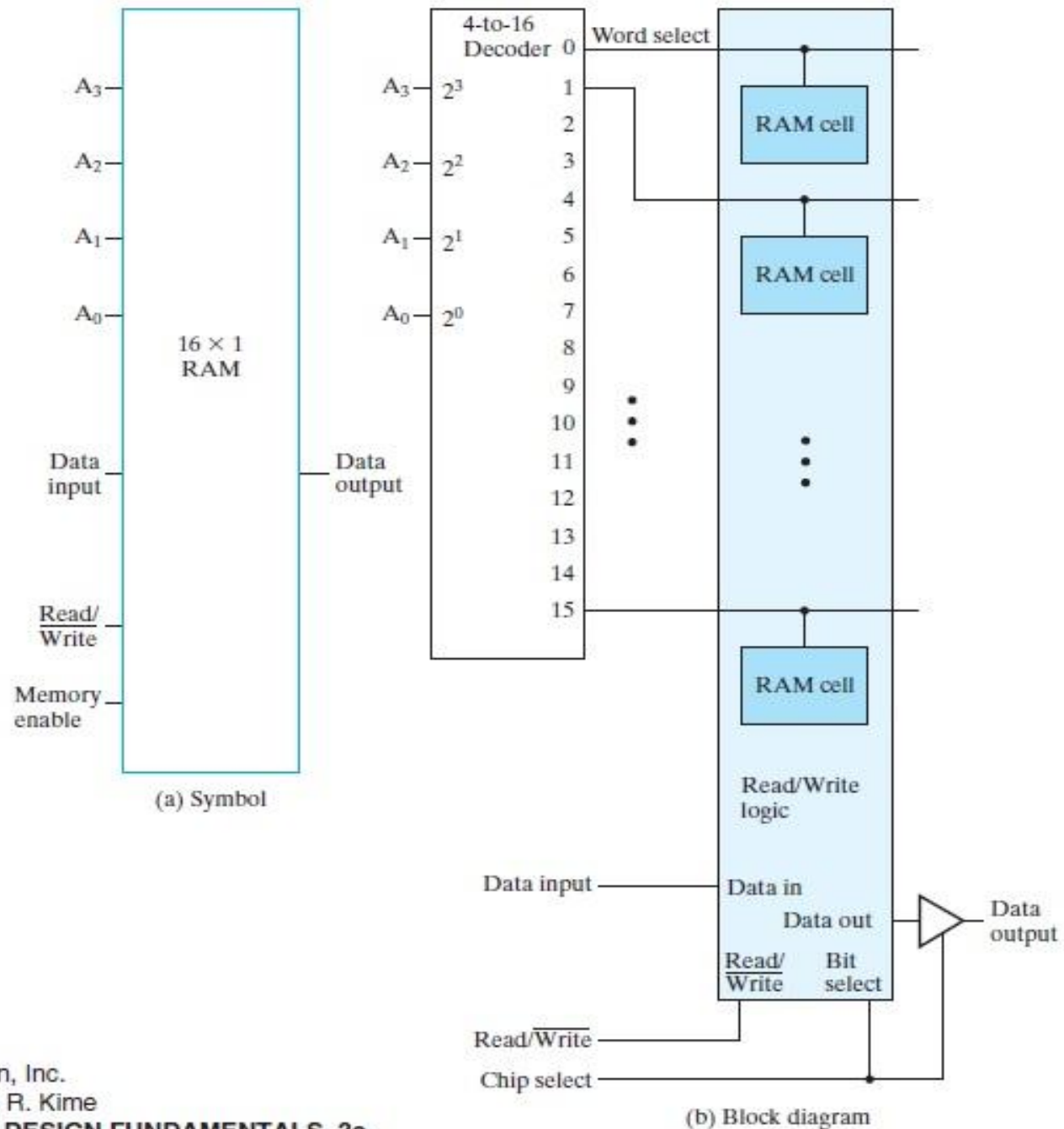
Random Access Memory (RAM)

- The data consists of n lines (for n -bit words). **Data input lines** provide the information to be stored (*written*) into the memory, while **data output lines** carry the information out (*read*) from the memory.
- The **address** consists of k lines which specify which word (among the 2^k words available) to be selected for reading or writing.
- The control lines *Read* and *Write* (usually combined into a single control line *Read/Write*) specifies the direction of transfer of the data.
- Example: A memory unit of 1024 words with 8 bit/word needs, 10 address line (i.e. address register must contain 10 FF) and buffer register must have 8 FF to store words transferred in and out of it.

Random Access Memory (RAM)

- Block diagram of a memory unit:





Random Access Memory (RAM)

- The **Write** operation:

- ❖ Transfers the **address** of the desired word to the address lines
- ❖ Transfers the **data bits** (the word) to be stored in memory to the data input lines
- ❖ Activates the *Write* control line (set *Read/Write* to 0)

- The **Read** operation:

- ❖ Transfers the address of the desired word to the address lines
- ❖ Activates the *Read* control line (set *Read/Write* to 1)



Random Access Memory (RAM)

- The Read/Write operation:

Memory Enable	<i>Read/Write</i>	Memory Operation
0	X	None
1	0	Write to selected word
1	1	Read from selected word

- Two types of RAM: Static and dynamic.
 - ❖ Static RAMs use flip-flops as the memory cells.
 - ❖ Dynamic RAMs use capacitor charges to represent data. Though simpler in circuitry, they have to be constantly refreshed.

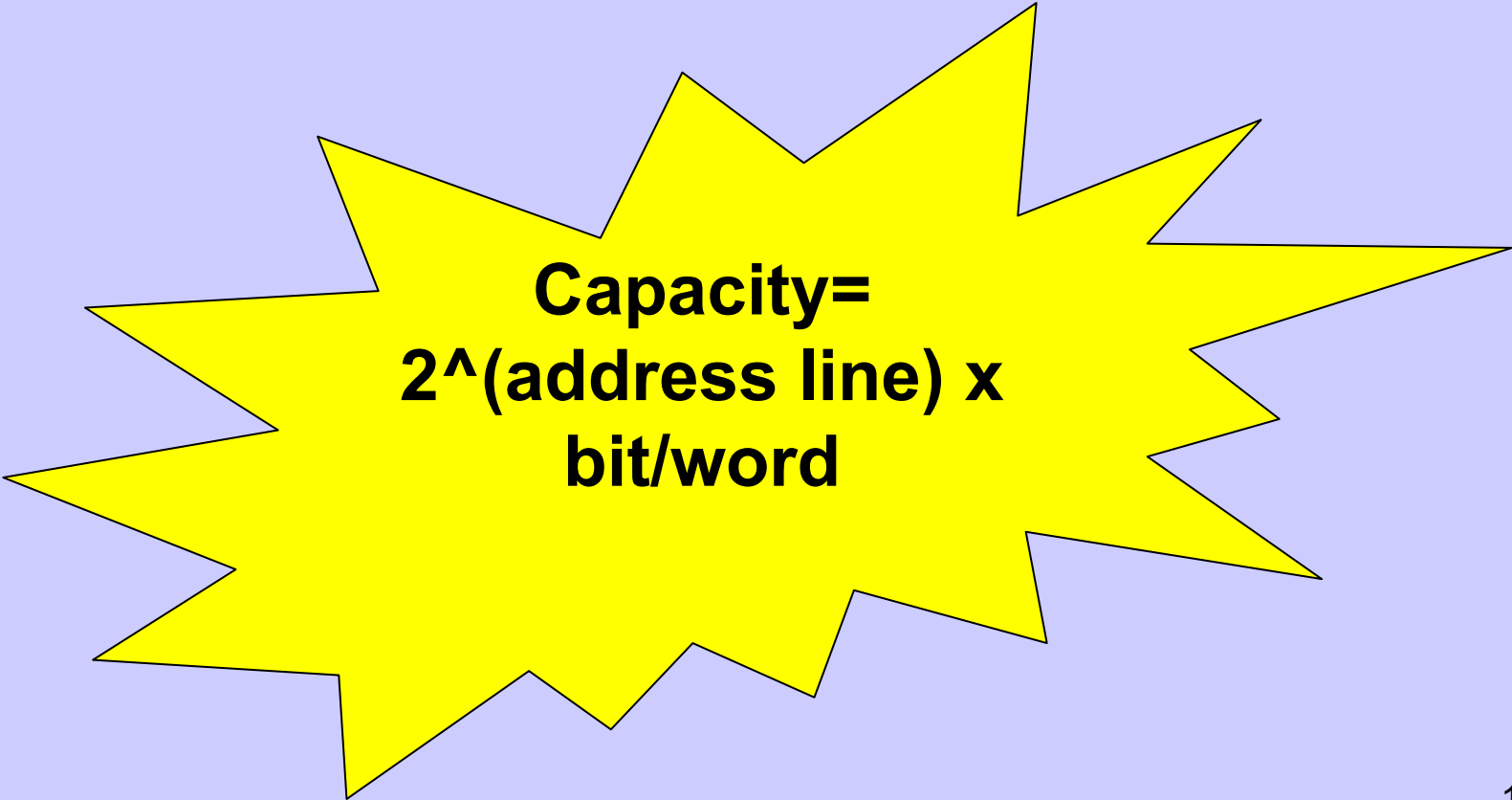
Memory Size

- Memory sizes might be specified in bits or byte(=8bit)
- Symbol: Bytes (B) and bits(b)
- Example, $2^{28} \text{ b} = 2^{28}/8 \text{ B} = 2^{28}/2^3 \text{ B} = 2^{25} \text{ B} = 2^5 * 2^{20} \text{ B} = 32 \text{ MB}$

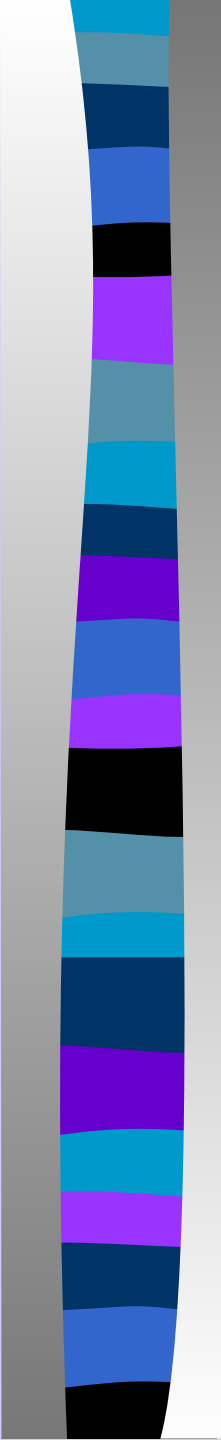
	Prefix	Base 2	Base 10
K	Kilo	$2^{10} = 1024$	$10^3 = 1,000$
M	Mega	$2^{20} = 1,048,576$	$10^6 = 1,000,000$
G	Giga	$2^{30} = 1,073,741,824$	$10^9 = 1,000,000,000$

2 formula to remember

- Capacity = no. of word x bit/word
- No. of word = $2^{\text{(address lines)}}$



**Capacity =
 $2^{\text{(address line)}}$ x
bit/word**



Test your self

- What is the total capacity of a $2^{16} \times 16$ memory:
 - A: 256 KB
 - B: 1 MB
 - C: 128 KB
 - D: 512 KB

- Solution: c

= $2^{16} \times 16$ bits/words

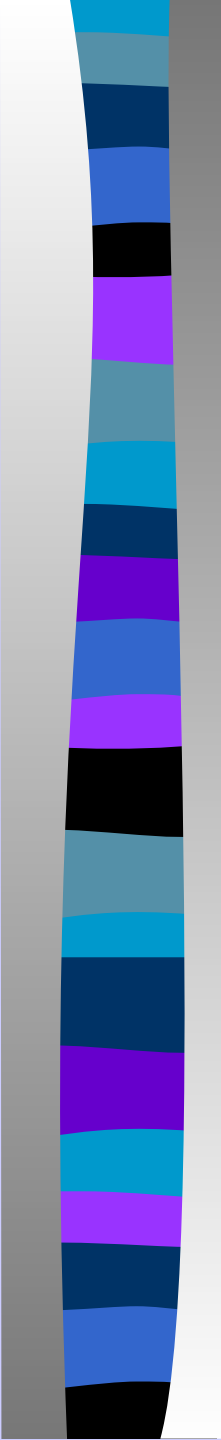
= 1048576 bits

= 1048576/8 bytes

= 131072 bytes

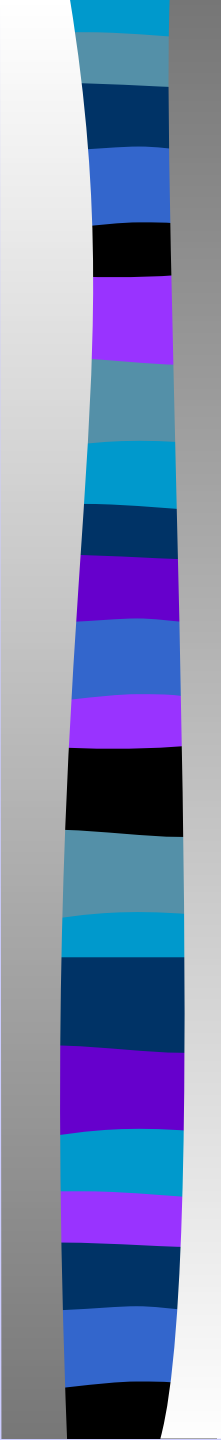
= 131072/1024 KB

= 128 KB



Test yourself

- How many address lines do we need for a 64 MB RAM with 32-bit words ?
 - A: 23
 - B: 20
 - C: 24
 - D: 21



Solution

Answer: C

Formula:

Capacity = no. of word x bit/word

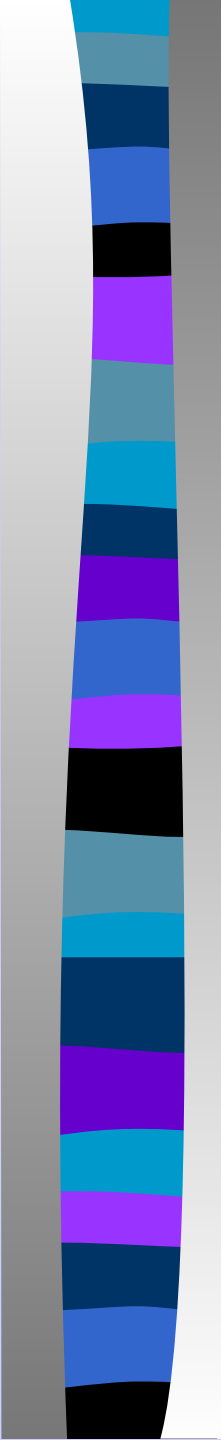
64MB = no. of word x 32bits/word

no. of word = 64MB / 32 bits/word

= $(64 \times 1024 \times 1024 \times 8) / 32$ [1 byte = 2^3 bits]

= 16777216 bits = 2^{24} bits

no. of address lines = 24

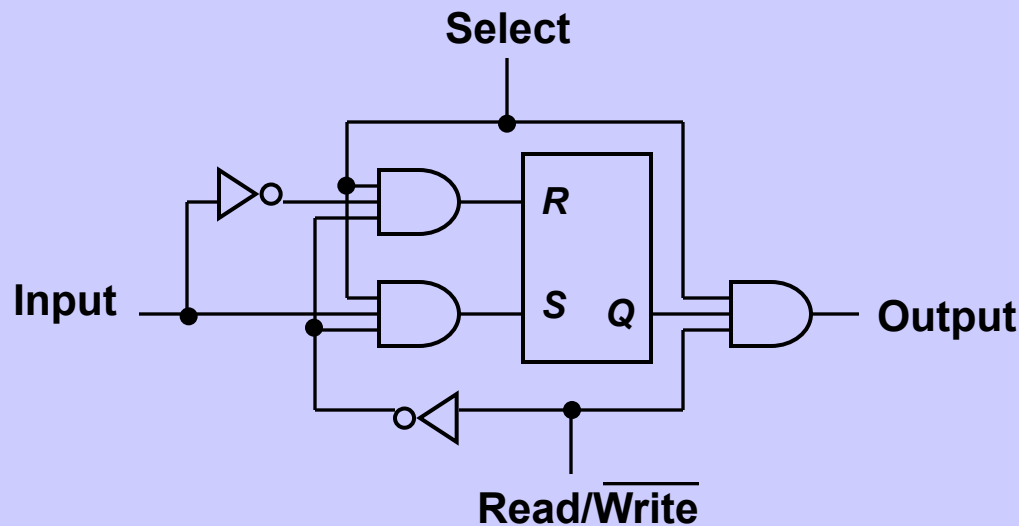


Binary cell

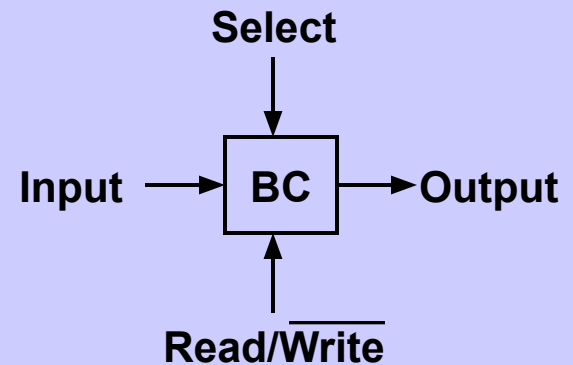
- Must be small so that many binary cell (BC) can fit in a small area
- Select input enable cell for reading/writing
- Read/write determines operation to be preformed on cell. A **1 in read/write** input forms a path to the output terminal. A **0 in read/write** input, information in the input terminal is transferred into the FF

Random Access Memory (RAM)

- A single memory cell of the static RAM has the following logic and block diagrams.



Logic diagram



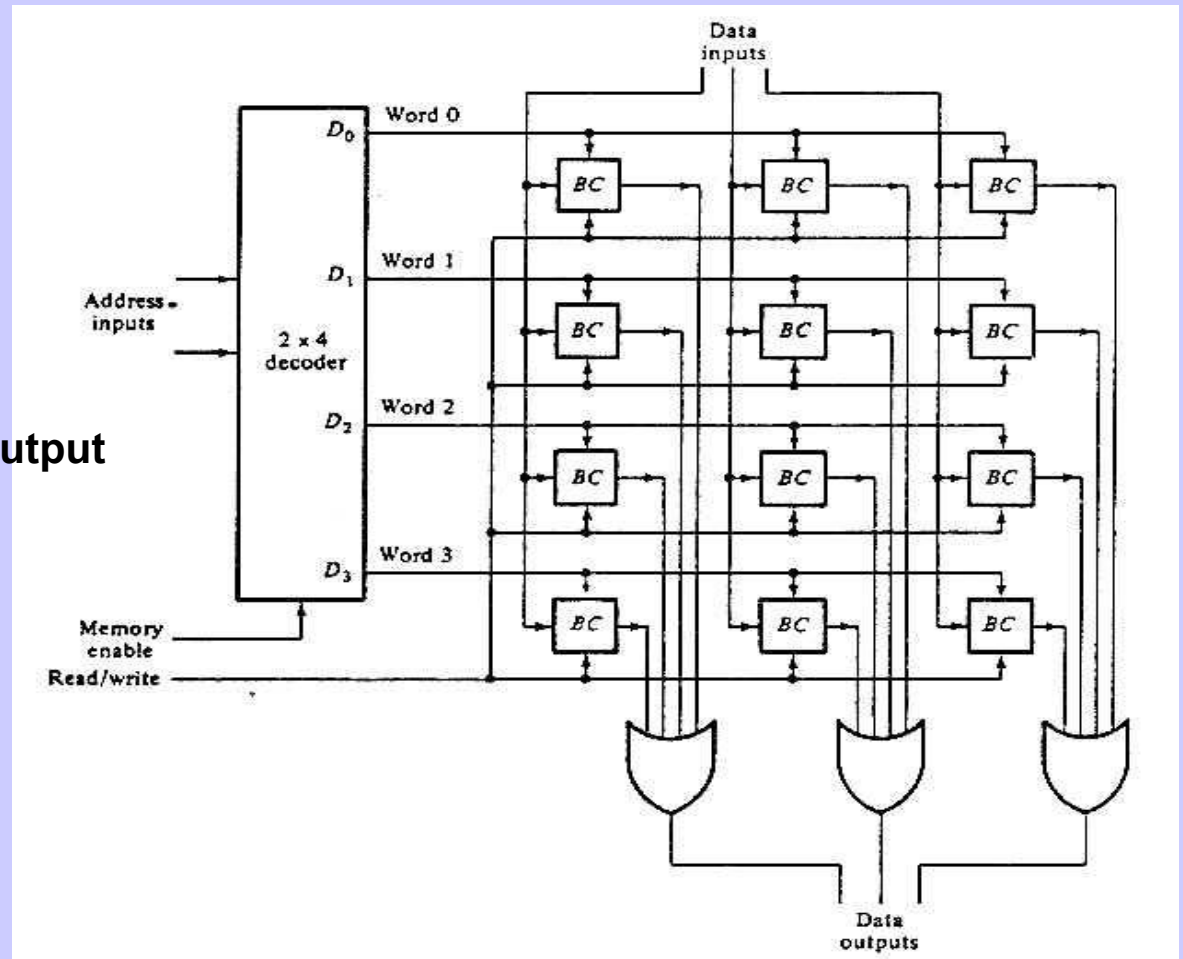
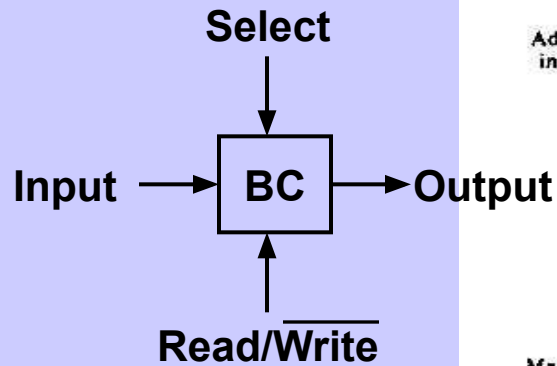
Block diagram

Random Access Memory (RAM)

of word

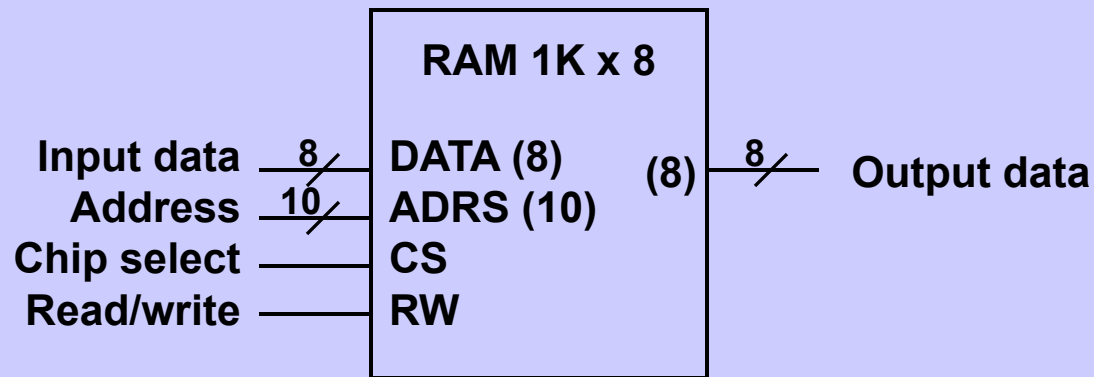
Bit/word

- Logic construction of a 4 x 3 RAM (with decoder and OR gates):



Random Access Memory (RAM)

- An array of RAM chips: memory chips are combined to form larger memory.
- A 1K x 8-bit RAM chip:



Block diagram of a 1K x 8 RAM chip